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A MAJOR PROJECT REPORT
ON

"LITMUS : A Multimodal Guardrail Framework for Jailbreak Detection in Vision-Language Models"

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LIST OF ABBREVIATIONS

ACL	: Annual Meeting of the Association for Computational Linguistics
AI	: Artificial Intelligence
BERT	: Bidirectional Encoder Representations from Transformers
CLIP	: Contrastive Language-Image Pre-training
DeBERTa	: Decoding-enhanced BERT with Disentangled Attention
MLP	: Multilayer Perceptron
NLP	: Natural Language Processing
OCR	: Optical Character Recognition
ViT	: Vision Transformer
VLM	: Vision-Language Model

CHAPTER 1

INTRODUCTION

1.1 Background

Safety alignment helps language models refuse harmful requests by using instruction tuning and preference optimization, which teach the model to give safer responses. However, this safety is learned behavior rather than a fixed rule, so it can sometimes be bypassed with unusual inputs that the model was not trained to handle. This is known as a jailbreak. To reduce this risk, companies use guardrails, which are extra safety systems that check user inputs before they reach the model and often check the model's outputs before they are shown to the user. Guardrails are useful because they work with different AI models, can be updated easily, and provide an additional layer of security.

The introduction of vision capabilities makes jailbreaks more challenging because attackers can exploit images in new ways. One method is typographic injection, where harmful instructions are written inside an image instead of the text prompt, allowing the vision model to read and follow them while text-only filters fail to detect them. Another method is query-relevant pairing, where the text prompt appears harmless, but a related image provides the harmful context, and the model combines both to understand the malicious intent. A third method is continuous-space adversarial perturbation, where attackers make tiny, almost invisible changes to an image's pixels that can reduce the model's tendency to refuse harmful requests. Since images contain continuous pixel values, they provide many more opportunities for such attacks than text, making vision-language models more difficult to secure.

1.2 Problem Statement

Vision-Language Models (VLMs) can be vulnerable to jailbreak attacks where harmful instructions are hidden in images or combined with seemingly harmless text. Existing text-based safety guardrails may fail to detect these attacks. Therefore, this project aims to develop a multimodal guardrail that analyzes both text and image inputs to detect and classify potential jailbreak attacks, improving the safety of vision-language systems.

1.3 Objectives

The key objectives of this project are:

1. To design and implement a lightweight multimodal guardrail that detects jailbreak attempts in combined text-and-image inputs before they reach the downstream language model.
2. To evaluate the guardrail against three major multimodal jailbreak failure modes and measure its effectiveness using suitable evaluation resources.

1.4 Scope and Delimitations

The scope of this project is to develop a lightweight multimodal guardrail that detects jailbreak attempts in combined text-and-image inputs before they reach a downstream language model. The system will focus on detecting typographic injection, query-relevant pairing, and adversarial image perturbation attacks and will classify inputs as safe or potentially malicious. The project is limited to text and image inputs and does not cover audio or video-based attacks. It focuses specifically on jailbreak detection rather than addressing all possible AI security threats, and the evaluation will be limited to selected datasets, attack types, and experimental conditions.

CHAPTER 2

LITERATURE REVIEW

The deployment of Vision-Language Models (VLMs) has introduced cross-modal capabilities alongside severe security risks from multimodal jailbreak attacks. This chapter reviews safety alignment limitations, multimodal attack taxonomies, existing defense paradigms, and the research gaps motivating this project.

2.1 Safety Alignment and Multimodal Jailbreak Taxonomies

Safety alignment methods such as Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO) steer model responses toward predefined safety guidelines [1, 2]. However, alignment forms a soft probabilistic barrier rather than a strict constraint, allowing adversaries to induce policy violations through jailbreak attacks [3, 4]. In VLMs, visual inputs expand the attack surface across three primary failure modes [2, 5]:

- **Typographic Injections**: Malicious textual instructions are rendered visually within an image, bypassing text-only safety filters while instructing the visual reasoning module [5, 6].
- **Query-Relevant Pairing**: Harmless text is paired with a visual prompt where neither is individually toxic, but their combined semantic synergy triggers harmful generation [5, 7].
- **Continuous-Space Adversarial Perturbations**: Gradient-optimized pixel perturbations disrupt internal safety representations in the vision encoder, evading refusal triggers [4, 8].

2.2 Existing Defense Paradigms and Guardrails

Existing defense frameworks can be classified into prompt-level text guardrails, internal state monitoring, and multimodal feature fusion filters:

- **Text-Centric Guardrails**: Frameworks such as *JailGuard* [3] and deep transformer classifiers [4] evaluate prompt mutations and text semantics. However, they are incapable of inspecting image contents or cross-modal context.

- **Internal Representation Defenses**: Methods like *HiddenDetect* [9] track transformer layer hidden states, Zhu et al. [8] apply null-space projection steering, and *JailDAM* [10] employs latent memory networks. While effective, they require white-box access and introduce heavy computational overhead, making them impractical for black-box API deployment.

- **Multimodal Fusion Guardrails**: Hua et al. [6] demonstrated that multi-branch representations combining raw image features, OCR text, and user text improve detection accuracy. Similarly, Liang et al. [7] showed that decoupled visual-textual encodings enhance generalization to unseen attacks.

2.3 Summary of Related Works and Research Gaps

Table 2.1 summarizes key existing literature in relation to the proposed system.

Table 2.1: Comparative summary of existing jailbreak detection approaches.

Framework	Modality	Core Strategy	Key Limitations
JailGuard [3]	Text	Mutation and response scoring	Blind to visual prompt injections
HiddenDetect [9]	Vision+Text	Layer hidden-state monitoring	Requires white-box access; high latency
Zhu et al. [8]	Vision+Text	Null-space activation steering	High inference cost; architecture dependent
Hua et al. [6]	Vision+Text	Multi-branch feature scoring	Lacks calibrated gating for input filtering
Proposed System	**Vision+Text**	**Tri-branch Encoder + Gated MLP**	**Lightweight pre-VLM input-level guardrail**

Identified Gaps and Justification: Conventional defenses either ignore visual vectors or depend on computationally expensive white-box activation monitoring. Furthermore, unimodal vision classifiers fail on typographic and cross-modal synergistic attacks. To address these gaps, the proposed work develops a lightweight, model-agnostic guardrail leveraging a tri-branch architecture to filter harmful inputs before reaching the downstream VLM.

CHAPTER 3

METHODOLOGY

3.1 System Components

The proposed system is a lightweight multimodal guardrail designed to detect jailbreak attempts in combined text-and-image inputs before they reach the protected Vision-Language Model (VLM). The methodology consists of several stages, including input routing, feature extraction, feature projection, feature fusion, classification, threshold-based decision making, and final interaction with the protected VLM.

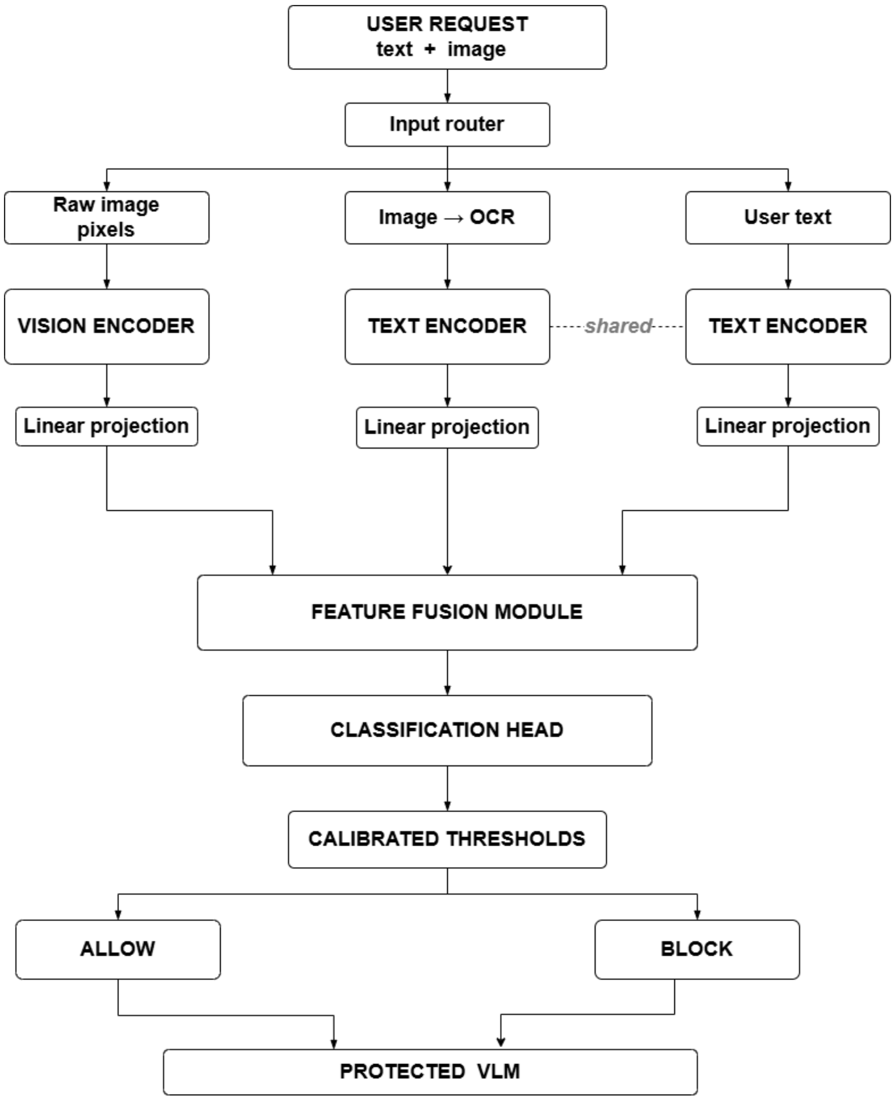

Figure 3.1: System Design Architecture for Multimodal Guardrail

3.1.1 User Request

The system first receives a user request containing text and an image. Both modalities are considered because a jailbreak instruction may be present directly in the user's text or hidden within the image.

3.1.2 Input Router

The input router receives the complete user request and separates it into three branches: raw image pixels, text extracted from the image using Optical Character Recognition (OCR), and the original user text. This separation allows the system to independently analyze the visual and textual information.

3.1.3 Raw Image Processing

The raw image pixels are directly passed to the vision encoder. This branch captures visual information from the image, including visual patterns, objects, layouts, and text-like features that may not be successfully extracted by OCR.

3.1.4 Vision Encoder

The raw image is processed using a vision encoder . The encoder converts the image into a numerical feature representation. These features contain information about the visual content of the input and are later used by the feature fusion module for jailbreak detection.

3.1.5 Image-to-OCR Processing

The image is also passed through an Optical Character Recognition (OCR) module to extract text present inside the image. This is important for detecting typographic injection attacks, where a malicious instruction is embedded as text within an image.

3.1.6 OCR Text Encoder

The text extracted by OCR is passed to a text encoder. The text encoder converts the extracted text into a numerical feature representation. These features provide semantic information about the text contained in the image.

3.1.7 User Text Encoder

The original user text is processed using text encoder. The encoder converts the user's textual request into a feature representation. The same encoder weights are shared between the OCR text and user text branches so that the system can process language consistently regardless of its source.

3.1.8 Linear Projection

The outputs of the vision and text encoders may have different dimensions. Therefore, linear projection layers are used to transform the extracted features into a common feature dimension. In the proposed architecture, the features are projected into a shared dimension.

This common representation allows the visual features, OCR text features, and user text features to be combined effectively in the feature fusion module.

3.1.9 Classification Head

The fused multimodal representation is passed to a classification head. The classification head is a neural network that produces a jailbreak risk score representing the likelihood that the input contains a malicious or jailbreak attempt.

For example, a lower risk score represents a safer input, while a higher risk score indicates a potentially malicious input.

3.1.10 Feature Fusion

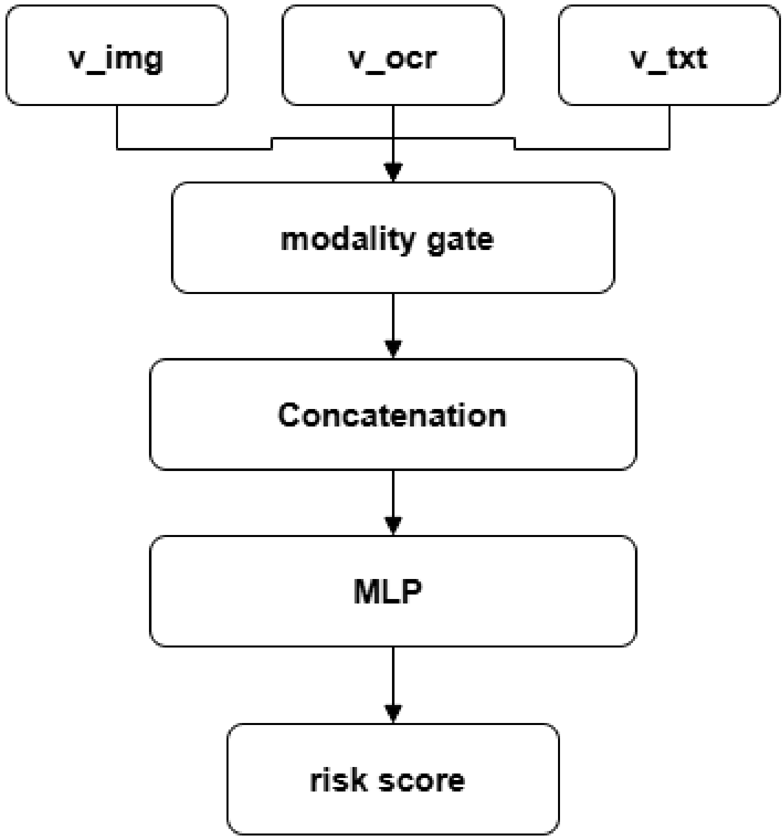

Figure 3.2: Feature Fusion Module Architecture

The baseline feature-fusion approach takes the feature representations from the image, OCR, and textual modalities as inputs [6]. The image representation captures visual information, the OCR representation captures text extracted from the image, and the textual representation captures information from the input prompt or request.

The three modality representations are first passed through a modality gating mechanism. The gate learns the relative importance of each modality and assigns appropriate weights to their corresponding feature representations. This allows the model to emphasize more informative modalities while reducing the influence of less relevant ones.

After the gating process, the resulting modality representations are concatenated to form a unified multimodal feature representation. This combined representation contains information from the image, OCR, and textual modalities and provides a common representation for subsequent processing.

The concatenated representation is then passed through a multilayer perceptron (MLP).

The MLP progressively transforms the fused representation into a compact feature representation while learning interactions among the different modalities. The resulting representation is finally used to produce the risk score of the input.

3.1.11 Calibrated Thresholds

The risk score is passed through calibrated decision thresholds. Two thresholds are used to divide the risk score into different decision regions. Based on the obtained risk score, the input is classified as *Allow* or *Block*.

3.1.12 Allow Decision

If the risk score is below the lower threshold, the input is classified as *Allow*. The request is considered sufficiently safe and is forwarded to the protected Vision-Language Model for further processing.

3.1.13 Block Decision

If the risk score is above the blocking threshold, the input is classified as *Block*. The request is considered potentially harmful or malicious and is rejected by the guardrail without being forwarded to the protected Vision-Language Model.

3.1.14 Protected Vision-Language Model

The protected VLM is the downstream model that processes the requests accepted by the guardrail. The guardrail acts as an additional security layer between the user and the VLM, preventing potentially malicious multimodal inputs from directly reaching the model.

CHAPTER 4

PROJECT WORK PLAN AND GANTT CHART

4.1 Project Execution Timeline

The project execution timeline spans a 5-month period from August 2026 to December 2026. The development milestones are structured across distinct technical phases, with project documentation and report writing conducted continuously from project initiation to final submission (Months M1 to M5).

4.2 Gantt Chart

The breakdown of tasks, execution periods, and development phases is illustrated in the Gantt Chart below:

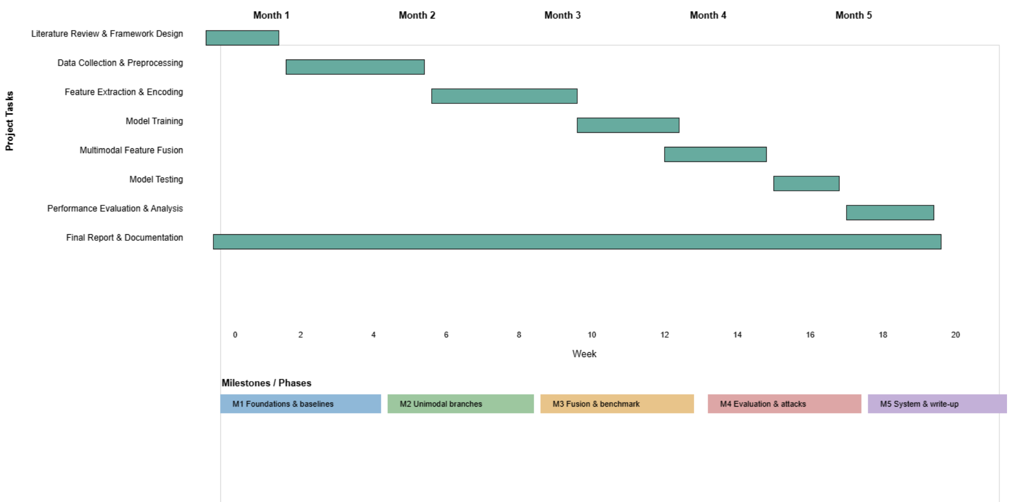

Figure 4.1: 5-Month Project Execution Gantt Chart (August 2026 – December 2026)

CHAPTER 5

EXPECTED RESULTS AND DISCUSSION

The proposed multimodal guardrail is expected to effectively detect jailbreak attempts by analyzing both text and image inputs. Each request will be classified as either *Allow* or *Block* based on its predicted risk score. The system is expected to detect attacks such as typographic injection, query-relevant pairing, and adversarial image inputs. Its performance will be evaluated using accuracy, precision, recall, F1-score, and false-positive rate. The expected outcome is a lightweight and reliable guardrail that blocks harmful requests while allowing legitimate requests to reach the protected Vision-Language Model.

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